ENTERPRISE SECURITY OPERATIONS PLATFORM

Seventh AI Vision

The complete feature inventory — AI video surveillance, security intelligence and guard workforce management in one multi-tenant platform, built for Singapore security agencies.

319	1,169	125	124	59	11
DATABASE TABLES	API OPERATIONS	API MODULES	WEB ROUTES	MOBILE SCREENS	AI DETECTION MODULES
228	**8**	**179**	**7,052**	**4**	**3**
PERMISSIONS	BUILT-IN ROLES	MIGRATIONS	AUTOMATED TESTS	LANGUAGES	CLIENT APPS

What this document is

Every capability listed here was read from the source — the API routers, the database migrations, the services and the client pages — not from a specification of what was intended. Where a feature has a documented limitation, this document says so rather than omitting it. What is still being built is listed separately, under *In progress and planned*, and is not counted in the figures above.

The platform is one system whose parts share a tenant, a permission model and an audit trail: AI-driven video surveillance, the security intelligence that relates what the cameras, drones, patrols and guards report, and the workforce management that turns an alert into someone actually attending. Each part runs without the others — a guarding-only customer needs no cameras, and a camera-only customer needs no roster — but the value is in the join.

One rule runs through all of it: AI detects, relates, assesses and recommends. An authorised person makes the security decision.

Edition 28 · generated 4 November 2026 · platform at migration 0179

About this edition

This is a living document. It is rebuilt from a script kept beside it in the repository (`scripts/make_feature_pdf.py`) each time the platform moves, so it is regenerated rather than hand-edited. A feature appears in the numbered modules only once it is merged, tested and running; until then it is in the table of what is in progress and planned.

Edition	Date	Platform at	What it covers
Edition 1	9 September 2026	0100	The first reference: thirteen modules, from identity to compliance.
Edition 2	7 October 2026	0148	Adds the vendor's platform console, Virtual Patrolling, Drone Patrol, AI Security Intelligence, and the first six phases of the enterprise expansion: investigation, evidence packages, the security map, guard response with SLA and escalation, the occurrence book reviewed and summarised, and the SOP library. Adds the sections on where AI is used, what is in progress, and known limits.
Edition 3	8 October 2026	0150	Adds phases 7 and 8 of the enterprise expansion: visitor and contractor authorisation (module 23), and device health, the asset register and maintenance work orders (module 24). The vendor's console becomes module 25.
Edition 4	8 October 2026	0151	Adds phase 9 of the enterprise expansion: risk patterns and advice (module 25) — where what went wrong gathers, what stands out with how much history it rests on, and a person's answer. The vendor's console becomes module 26.
Edition 5	8 October 2026	0155	Adds phases 10, 11 and 12 of the enterprise expansion: the operations board, the daily briefing and nine operations reports (module 26); workforce readings and recommendations (module 27); and security cases (module 28). The vendor's console becomes module 29.
Edition 6	9 October 2026	0156	Adds phase 13 of the enterprise expansion: data retention and subject reports (module 29), with a security sweep over everything the expansion added. Phase 14 follows eleven chains end to end as tests. The fourteen phases are built; the vendor's console becomes module 30.
Edition 7	9 October 2026	0158	The seven things left for the owner were decided, and are carried out: the occurrence book cannot be rewritten by the application; exports write formula-like text as text; the data-subject export gives its subject's own and is audited; privacy masks are read with a credential; four kinds of the newer records may be given a retention period; and a badge used outside its authorisation may be handed to the intelligence layer. The vendor's licence catalogue gains the fourteen modules built since it was written, and Manage Licenses gains Drone Patrol's own licence. The phone gains four screens; the desktop app is 1.0.5. **Corrects module 02:** earlier editions said privacy zones are masked out of live view and recordings. They are recorded for a camera, and nothing applies them yet.
Edition 8	9 October 2026	0158	Privacy zones are applied. The owner decided it the day edition 7 said they were not: what is under a zone is painted out of the live view, of what the AI is given, of recordings and of the images a patrol keeps, and a third tab on the Zones page draws them. Module 02 says what is now true, with its limits. The desktop app is 1.0.6.
Edition 9	10 October 2026	0158	Privacy zones on the phone: whoever manages privacy draws one from a camera's live view, sees that camera's zones and deletes one, each after the phone has asked; a masked camera is labelled on the phone's live view and live wall. No route is added. The phone has 57 screens.
Edition 10	10 October 2026	0158	The platform owner's two-factor sign-in now completes on the web, the phone and the desktop app. The server had asked for the code since the control was switched on; until this day no client had a step to give it. The desktop app is 1.0.7. The sign-in page is no longer reused by a browser without asking.
Edition 11	14 October 2026	0158	How the web console moves, waits and fails, in a section of its own: every page has its own entrance and its own loading; a page that could not fetch says so; a camera's tile says what it is doing; nothing is drawn as live that is not. No route and no table is added. **Corrects Known limits:** the sign-in page has fetched a newer console by itself since 10 October.

Edition	Date	Platform at	What it covers
Edition 12	14 October 2026	0158	The Windows app is 1.0.8 and carries all of the web console's new motion and loading. Every screen of the phone has its own entrance and its own loading, with the same bounds. No route and no table is added.
Edition 13	14 October 2026	0158	The phone keeps the web console's loading rules: a screen says when it failed and can try again; what arrives is marked; a tick only once the server has said so. No route and no table is added.
Edition 14	15 October 2026	0162	Adds module 30, fleet tracking and geofences: trackers through Traccar, the live map, journeys played back, and geofences a wobbling fix does not trip. The vendor's console becomes module 31.
Edition 15	23 October 2026	0166	Adds module 31, Footage Review: uploaded video searched by description, with events and possible altercations for a person to judge, kept 30 days. The vendor's console becomes module 32. **Corrects** what is said of language models.
Edition 16	23 October 2026	0166	Windows app 1.0.9. A real S3 bucket is reached with the pods' role (module 31).
Edition 17	23 October 2026	0167	Face registration, and each check-in selfie matched to it (module 06).
Edition 18	26 October 2026	0174	Module 32, the AI assistant and learning; overtime approval.
Edition 19	28 October 2026	0174	The small free model runs, helped (module 32); Windows 1.0.12.
Edition 20	31 October 2026	0174	The AI button on every phone screen, and the AI's decisions pop up on the web, the desktop app and the phone (module 32).
Edition 21	31 October 2026	0174	Windows 1.0.13 carries the decision card.
Edition 22	31 October 2026	0174	Windows 1.0.14; the web, the desktop app and the phone say which build is running.
Edition 23	1 November 2026	0174	Windows 1.0.15; the assistant sees only the person's own organisation, and the free model is sent less.
Edition 24	3 November 2026	0175	Singapore payroll (module 11): CPF by date, PR rates, bonus ceiling, MOM's PWM floors.
Edition 25	3 November 2026	0176	Retention for the history of modules 20 to 26; evidence packages removed by a person (18, 29).
Edition 26	4 November 2026	0177	Cases (module 28): people told when put on one; overdue, quiet and waiting cases escalate.
Edition 27	4 November 2026	0178	Module 26: the daily briefing emailed when published; the nine reports on a schedule as CSV and PDF.
Edition 28	4 November 2026	0179	Drone Patrol (module 15): a drone's whole picture withheld while it is at a privacy area.

How the document is laid out

- **At a glance** — one page for the person who signs.
- **Part 1, modules 01 to 13** — the platform as the first edition described it, brought up to date.
- **Part 2, modules 14 to 33** — what has been added since: patrols from a screen and from the air, security intelligence, the enterprise expansion — from investigation to cases — the fleet, footage review, AI.
- **Where AI is used, and where a person decides** — every place a model or a rule does work, what it is trusted with, and what stays a person's. **AI capabilities added or upgraded later are recorded here.**
- **In progress and planned** — what the owner decided and how each now stands, and what waits on something outside the code.
- **Client applications, roles, Singapore, and known limits.**

At a glance

A security agency runs on four or five systems that do not talk to each other: a spreadsheet roster, a messaging group for incidents, a paper occurrence book at every guardhouse, a payroll spreadsheet, and a filing cabinet for whatever a client asks about later. This is one system that replaces all of them.

What changes

	Today	With Seventh AI Vision
Client asks for proof of a patrol	Phone calls, a photo of a page	On screen in seconds — or the client checks their own login
A key is unaccounted for	Nobody knows until it is needed	Shown at every handover, with who took it and when
Leave approved on a live roster	Someone remembers, or nobody does	The uncovered post queues until it is filled
Month-end payroll	Hours retyped from timesheets	From real check-ins, supervisor-approved first
A lone officer collapses	Nothing, until somebody notices	Man-down alerts the control room with their last position
Invoicing a client	Estimated from the contract	From the hours actually worked at their site
Five alerts from one event	Five things to read, on five screens	One situation, with what is unusual about it and a suggested response
"Where was that vehicle seen?"	Somebody scrubs through recordings	Every sighting of the plate, in order, with time and distance between
Evidence asked for months later	Whatever has not been purged	A sealed package under a hold, with its chain of custody
A guard is sent to an incident	A call, and hoping	Accepted, on the way, arrived, reported — each recorded, and late ones escalated

The platform in thirty-three lines

- **Identity and access** — 8 roles, 228 permissions, SSO, 2FA, audit log, per-tenant branding, four languages.
- **Video surveillance** — cameras, live wall, recording with checksums, playback, privacy zones painted out of the picture, NVR.
- **AI analytics** — eleven detection modules, licensed individually.
- **Alerts and incidents** — triage, dispatch, escalation, emergency broadcast.
- **Roster** — a whole month auto-scheduled against nine rules, reviewed as a draft before it goes live.
- **Attendance** — face check-in with liveness, geofenced, eight live states.
- **Guard operations** — patrol scanning, occurrence book, post orders, panic button, man down.
- **Guardhouse registers** — keys, lost & found, defects, equipment, uniform, and a handover two people sign.
- **Visitor management** — pre-registration, passes, contractors, deliveries.
- **Physical security and IoT** — access control, alarms, barriers, parking.
- **HR and payroll** — leave, training, timesheets, CPF, overtime, public holidays, PWM grades and wage floors, IR8A.
- **Commercial** — invoices from real hours, and a client portal.
- **Compliance** — chain of custody, PDPA, retention, scheduled reports.

- **Virtual Patrolling** — a scheduled camera-by-camera inspection from a screen, with a report.
- **Drone Patrol** — planned flights whose camera the AI watches like any other.
- **Footage Review** — uploaded video searched by description, events on a timeline, possible altercations for a person to judge; kept 30 days.
- **AI Security Intelligence** — events related into situations, assessed, with a suggestion a person decides on.
- **Smart Investigation** — one search across fourteen sources, and the trail of a plate or a watchlist face.
- **Evidence packages** — sealed, checksummed, held against deletion, exported.
- **Security map** — cameras, guards, drones, incidents and the places of a site on one map.
- **Guard response** — the steps of a response, three SLA clocks, escalation policies.
- **Occurrence book, reviewed** — search, supervisor review, correction, standing instructions, a shift summary.
- **SOP library** — versioned, approved procedures, quoted beside the incident.
- **Visitor authorisation** — a host's yes or no to a visit, for where and for how long, read at the gate.
- **Assets and maintenance** — device health as devices report it, an asset register, and work orders a person accepts.
- **Risk patterns** — where what went wrong gathers, by hour, day and place, with how much history each statement rests on.
- **Operations board and briefing** — every part of the operation counted together, a daily briefing a person publishes, and nine reports as files.
- **Workforce readings** — what is recorded of a guard's work, beside how much there was to do, and recommendations a manager answers.
- **Cases** — a matter held together: its people, tasks, linked records and names, closed by two.
- **Data retention** — every retention period in one statement, and where a person appears in the records.
- **Fleet** — vehicles, Traccar trackers, journeys played back, and geofences a wobbling fix does not trip.
- **AI assistant and learning** — ask in your own language, changes wait on Confirm; twelve features learn from the organisation's own work.
- **Platform console** — the vendor's own: customers, pricing, invoicing, errors, support sessions.

Built for Singapore, not adapted to it

PLRD licence expiry per officer, and what a site requires an officer to hold. PWM grades across the seven-grade ladder, with wage floors pay is checked against. CPF from CPF Board's tables, PRs' first two years included. Gazetted public holidays and holiday pay. The 72-hour monthly overtime cap flagged on every run. IR8A. PDPA consent and data subject requests. An occurrence book built against the Private Security Industry Act. SGD invoicing. English, Chinese, Malay and Tamil. Runs hosted or on your own servers.

Part 1 — Platform capabilities

The thirteen modules of the first edition, brought up to date. Where a module has grown into one of its own since, it says where to read on.

01. Multi-tenancy, identity and access

One installation serves many security agencies, isolated in the database rather than in application code.

- **Row-level security on every table.** Tenant isolation is a Postgres policy, forced even on the table owner, not a WHERE clause a developer can forget. Records that must not be rewritten are granted insert and read only, and the application's database role cannot empty a table.
- **Eight built-in roles** — Super Admin, Admin, Manager, Supervisor, Operator, Security Guard, Viewer, Client — plus custom roles defined per tenant, across 228 permission codes.
- **Site scope.** A user can be held to particular sites; every list, search and record honours it.
- **The platform operator is not a customer's administrator.** Super Admin runs the platform from a tenant of its own and holds none of a customer's day-to-day permissions (module 33).
- **Single sign-on and SCIM** for automated user provisioning and de-provisioning from the customer's own directory.
- **Two-factor authentication (TOTP)** — required, not optional, for the platform owner — with IP allow-listing, API keys, rate limits, and per-user session management with forced revocation and account unlock. After a correct password the server asks for the 6-digit code, and the web, the desktop app and the phone each have the step that gives it; a locked account is told so in the server's words. *Corrected in edition 10:* until 10 October 2026 no client had that step, so an account with two-factor switched on could not be signed in to at all.
- **Hash-chained audit log** covering every privileged action, with the actor, role, tenant, site, result and request id, and verifiable end to end.
- **Per-tenant branding** — company name, logo and primary colour — with the product name held separately so white-labelling never hides which platform is running.
- **Per-tenant timezone** applied to every date the application renders, so a Singapore operator never reads a UTC timestamp.
- **Four languages:** English, Chinese, Malay and Tamil.

02. Video surveillance

Cameras, streams and recordings, with the operator surfaces built around watching many at once.

- **Camera management** with ONVIF discovery, PTZ control, and RTSP credential validation before a camera is saved rather than after it fails.
- **Live Wall** — a multi-camera video wall with saved layouts and profiles, and genuine pop-out windows for multi-screen control rooms.
- **Continuous recording** with retention policies that inherit tenant to site to camera, so an exception is set once where it applies.
- **Recording integrity.** Each recording carries a checksum that is verified, and the platform's health check reports how many streams should be recording against how many are.
- **Playback and a recordings library**, with export.
- **Privacy zones, applied.** A zone drawn on a camera's picture is painted out of the live view, of what the AI is given, of recordings and of the images a patrol keeps, within about ten seconds. It is painted into the frame, not drawn over the video, so there is no unmasked copy to be sent or kept — and what is recorded after it

cannot be unmasked. If a camera's zones cannot be read, its picture is not shown. A zone is drawn and deleted by a signed-in person who manages privacy, on the Zones page, and each is one line in the audit log. *Added in edition 8.* **On the phone** (edition 9) the same person draws a zone from a camera's live view, sees that camera's zones and deletes one; the phone says what a zone does and asks once more before it masks, and asks before it deletes. A masked camera is labelled on the phone's live view and live wall. *Documented limitations:* a zone is fixed to the picture, not to the scene, so a camera that is turned moves out from under it; footage recorded before a zone was drawn is unchanged; a camera with a zone has no HLS view and is shown through the masked one; a recorder at the site that records the camera itself is outside it. A drone's picture is withheld whole at a privacy area instead (module 15).

- **NVR integration** and HLS streaming for browser playback.
- **Site map view** plotting sites with live camera health and manning (the full security map is module 19).
- **Detection heat maps** and trend analytics per camera and site.

03. AI analytics

Eleven detection modules, licensed individually per tenant so a customer pays for what they run.

- **Licence plate recognition (LPR/ANPR)** — entry and exit lanes, watchlist matching, vehicle journeys.
- **Face recognition** with watchlists and pgvector similarity matching.
- **Intrusion detection** against drawn restricted zones.
- **PPE compliance** — helmet, vest and equipment checks.
- **Crowd density** and zone occupancy.
- **Fire and smoke detection.**
- **Weapon detection.**
- **Behaviour analysis** for anomalous movement.
- **Camera tampering** — obstruction, defocus and repositioning.
- **Abandoned object detection.**
- **Fall detection** for slips and collapses in camera view.
- The same detectors watch a drone's camera (module 15), and what they report is related and assessed by the intelligence layer (module 16).

04. Alerts, incidents and response

Turning a detection into a decision, and a decision into an attended response.

- **Alert rules engine** with per-module thresholds, schedules and site scoping.
- **Deduplication rules** so one event does not become forty alerts.
- **Alert routing** to the right people by role, site and severity.
- **Alerts that are not about a camera** — a contractor's permit expiring, a visitor overstaying, a guard pressing panic — carry their own site, so a guarding-only agency with no cameras receives them, and they reach the people responsible for that site.
- **Incident management** — assignment, notes, severity, status and resolution, with bulk actions. A matter that takes more than one record — investigators, tasks, linked evidence, a closing by two — is a case, module 28.
- **Command Centre** — the live operational picture across every site.
- **Action Centre** — cross-role duty guidance telling each person what needs them next.
- **Emergency broadcast** to all staff, a role or a site, with per-recipient acknowledgement tracking.
- **Dispatch** for sending a named guard and tracking arrival. The guard's own steps, the SLA clocks and escalation are module 20.

- **Notification channels and rules** — email, SMS, webhook, in-app and push, with a delivery log.

05. Roster and scheduling

Building the roster, and keeping it true when reality interferes.

- **Shift patterns** per site, guard and weekday, expanded into concrete shifts on a rolling window.
- **Auto-scheduler** scoring candidates against nine rules: minimum rest (11 hours), maximum consecutive days (6), day/night balance, stated preference, approved leave, the site's own duty team, minimum coverage, supervisor presence, and off-day stagger.
- **Per-shift-type duty strength.** A site is staffed separately for day and night duty, and the scheduler fills every post — not one guard and a warning.
- **Duty Teams** — named day and night teams per site, with the shortest-staffed sites sorted first. The scheduler draws from the team, falling back to the wider pool rather than leaving a post empty, and flags when it had to.
- **One post per guard.** An Operator or Security Guard holds one posting; Supervisors and Managers may cover several sites. The database holds the rule, so the same officer cannot be put on three sites at once.
- **What a site requires an officer to hold.** A tenant-wide baseline (every officer holds a valid licence) plus what a particular client's contract adds. An officer who does not meet it is reported daily — it is not a block, because a manager covering a 2 am no-show must not be stopped by the roster tool.
- **Guard shift preferences** — preferred shift type and off days, set on the guard's own record and honoured by the scheduler.
- **Draft rosters** reviewed and edited before publishing, with warnings for unfilled posts, coverage shortfalls and missing supervisors. Drafts are listed until published or discarded.
- **Planned from shift definitions** where a site has no patterns: its day and night shifts and strengths, in the organisation's own time zone. Days already past are skipped, and a roster that cannot be planned says why.
- **Schedules a whole calendar month in one action**, or any range up to 62 days, with the rules configured per run rather than read out of the source.
- **Cover requests.** When approved leave lands on a published shift, the uncovered post joins a queue that stays open until a supervisor assigns a replacement or records that none is needed. The system refuses a replacement who is on leave or already rostered on an overlapping shift.

06. Attendance

Proving a guard was where they were rostered, when they were rostered.

- **Selfie check-in and check-out** with liveness scoring to reject a photograph of a photograph. A guard checks in and out of their own shift.
- **Face registration** — each person registers their face once on the phone, with their agreement recorded; a supervisor changes a guard's, an admin a supervisor's. Every check-in selfie is compared with it. A mismatch never blocks the check-in: it is listed for a supervisor, beside the photo on file. A closed account's face is deleted that night.
- **Anti-spoof location** — mock-GPS detection flagged on the record.
- **Site geofence** as a radius or a drawn polygon, with off-site check-ins marked rather than silently accepted.
- **Per-site late grace** — a remote gate with one bus an hour cannot hold the same standard as a lobby on a train line.
- **Live attendance board** with eight states: on time, late, on break, due now, not reported, upcoming shift, approved leave and checked out.
- **Escalation ordering** — sites with guards due and nobody reported sort to the top under a no-guard-on-site banner.

- **Correction queue** for supervisor-approved amendments.
- **A shift left open is closed by the system** a set number of hours after check-in (15 unless the admin sets otherwise), with a reminder to the guard 15 minutes before; the record says the system closed it. A check-out away from the site is allowed and flagged.

07. Guard operations

What a guard actually does on shift, and the record it leaves.

- **Guard tours** — routes and checkpoints scanned by QR or NFC, with tour schedules and compliance reporting.
- **Digital Occurrence Book** — entries with immutable timestamps, built against the Private Security Industry Act requirement. Fifteen kinds of entry. Search, supervisor review and correction are module 21.
- **Post orders** per site, with per-guard acknowledgement so there is a record of who read what. Versioned, approved procedures are module 22.
- **Violations** raised against a guard, with review.
- **What the phone offers a guard is what the server allows:** their own patrols started and finished, a note on an incident, a visitor pre-registered. A button the server would refuse is not shown.
- **GPS tracking** and geofence entry/exit events. The fleet's vehicles, trackers, journeys and geofences are module 30.
- **SOS panic button** — writes a critical occurrence entry, raises an incident, and pushes a real-time alert to every supervisor watching.
- **Body-worn cameras** — assignment, recordings and events.
- **Man down.** The guard's phone watches its accelerometer and raises an alert when they stop moving, or are knocked and then lie still. A countdown they can cancel avoids a phone left on a desk waking the control room — but the deadline is held on the server, so a handset that shattered on impact, ran flat or lost signal still gets help sent. Off by default; the thresholds are a tenant setting.

08. The guardhouse registers

The books kept on paper at almost every site — the ones a client audit asks to see first, and the ones nobody can report on.

- **Key register.** A cabinet per site, and a log of who took which key, when, and whether it came back. Built so the useful question — what is still out right now — is answered directly, and the database refuses to let one key be issued twice. Keys go to contractors and tenants as often as to staff, so a holder can be a named person without an account.
- **Lost and found.** What was handed in, where it is stored, and who took it away. Only the last few characters of a claimant's identity document are ever stored — the check that matters is against the card in their hand. Property held past the retention period is reported, because an item kept indefinitely is a PDPA problem accruing quietly.
- **Facility defect log.** The blown stairwell light, the leak in the car park. The status ladder stops where a security company's control does: the record carries who it was referred to and the building's own ticket number, so chasing it is possible. Safety hazards and anything open past a fortnight are surfaced separately.
- **Equipment register.** Radios, torches, batons and body cameras signed out to a named officer. The condition it comes back in writes through to the item, so the next officer is not issued a broken torch.
- **Uniform register.** Issued by quantity and size, with partial returns — a guard who resigns hands back three of four shirts — and the deposit held against the kit.

- **One question answers both:** what is this officer holding? The screen a supervisor opens on somebody's last day.
- **Structured shift handover.** Two signatures, not one person's note. The outgoing guard completes a checklist configured per site; the incoming guard accepts it or DISPUTES it. A counted item is measured against the register, so eleven keys against a handover claiming twelve is visible rather than signed for. Every handover carries the keys still out, property held, kit signed out and defects open.

09. Visitor management (VMS)

The gatehouse: every visitor, not just the ones in vehicles.

- **All visit types** — walk-in, delivery, drop-off and pick-up — on one board.
- **Per-site form configuration.** Each site defines its own fields, types and required flags.
- **Pre-registration** as a dialog on the same page, so the gatehouse never navigates away mid-queue.
- **Fast LPR entry** — the plate is read and the record opened with known details already filled, because a vehicle at the barrier blocks the road while anyone types.
- **Visitor labels with QR**, 62 mm, printed at the operator's discretion rather than automatically.
- **QR check-out** — scanning the label closes that visit.
- **Exit-camera auto check-out** from an LPR read.
- **Automatic day-close** so a visit never runs past its own day.
- **Full action log** naming the operator behind every entry and exit.
- **Contractors** — vetting, accreditation, work permits with approval and a safety briefing, and deliveries.
- Host approval of a visit, the places it is authorised for, escort, and what the gate is told are module 23.

10. Physical security and IoT

The hardware around the guard.

- **Access control** — doors, credentials, rules and an event log.
- **Barrier control** with commanded open/close and an audit of who commanded it.
- **Alarm panels and zones** with event history.
- **IoT sensors** — readings, thresholds and alerts.
- **Car park management** — car parks, bays, zones, tariffs, sessions and dedicated LPR cameras, with a free-parking allowance per site.
- **Device protocol support** for third-party integration.

11. HR, leave, training and payroll

The workforce behind the roster.

- **Employee records** — NRIC/FIN, date of birth, nationality, work pass type and expiry, address, emergency contact, bank details and designation.
- **Document management** with expiry tracking and expiring-soon warnings on passports, work passes and certifications.
- **Leave** — configurable types, entitlement balances, requests with supporting documents, and approval that feeds the roster directly.
- **Training** — courses, a real quiz engine with question banks, attempts and pass records.
- **Certifications** with issuing body, number and expiry.

- **Payroll** — CPF from CPF Board's published tables for the payroll month: the age band, a PR's graduated first two years, the wage bands, CPF's rounding and the year's Ordinary Wage ceiling. Bonuses are Additional Wages up to the year's ceiling, then Recalculate. A new employee is asked Singaporean, PR or foreigner first, and the salary shows its CPF. Overtime at 1.5x; monthly, daily or hourly pay bases.
- **Timesheet approval.** A human between a check-in scan and a payslip. Hours are snapshotted when submitted, so an approval stays a signature on a fixed document rather than on one that keeps changing underneath. Payroll always reports which guards have no approved timesheet, and can be set to refuse to pay them.
- **Public holidays and shift allowances.** A gazetted holiday calendar per tenant, holiday pay, and named allowances attached to a shift type.
- **Overtime cap.** Hours beyond the 72-hour monthly statutory limit are flagged on the run rather than paid silently.
- **Overtime asked and approved.** A guard asks from the phone to work past the shift; a supervisor approves, or refuses with a note. Only approved overtime is paid. Payroll runs already made are not recalculated.
- **PWM grade and wage floors.** The grade is on the officer record across the seven-grade ladder, and pay is checked against a floor: MOM's 2024-2028 schedules, outsourced or in-house, full- or part-time, or a company's own, which may only be higher. A floor is judged as of the payroll period, so restating March is judged against March. Where no rate has been loaded the result is *not assessed*, never *compliant*.
- **PLRD licence number and expiry** tracked per officer.
- **Payslip PDFs** and an **IR8A** annual summary and PDF, and a year-end check of the bonus CPF ceiling.

12. Commercial

Getting paid for the hours actually worked.

- **Client management** with contacts and billing addresses.
- **Invoices generated from real attendance** — the loop closes from roster to check-in to worked hours to invoice line, per site.
- **Per-site bill rates** with overtime, tax and invoice PDFs.
- **Client portal** — the customer signs in and sees their own sites, incidents and invoices, and nothing else.
- **What the vendor charges the agency** — plans, module prices, subscriptions and the vendor's own invoices — is module 33.

13. Compliance, evidence and reporting

The record that survives an audit.

- **Evidence management** with an access log recording every view and download. Sealed packages, holds and the full chain of custody are module 18.
- **PDPA** — consent records and data subject access requests, with erasure and subject export.
- **Contractor accreditation** and work permit tracking, with expiry warnings.
- **Data retention policies** per tenant, site and camera, and a hold that stops a purge. Every period in force said in one statement, and a report of where a person appears in the newer records, are module 29.
- **Scheduled reports** delivered on a recurring basis by email or webhook.
- **Checksums on what is handed over** — patrol snapshots, patrol reports, recordings and evidence files — so a document can be checked later against what was written.
- **Exports** in operational formats, and a full analytics suite. The operations board, the daily briefing and the operations reports are module 26.

Part 2 — Added since the first edition

Eighteen modules built after the first edition. Each was added beside what already worked: new tables, new API modules and new screens, with the existing ones left as they were. Every one keeps the same rule — the platform detects, relates, assesses and recommends, and a person decides.

ADDED SINCE EDITION 1 · MIGRATIONS 0116, 0117, 0121

14. Virtual Patrolling

A scheduled, camera-by-camera inspection carried out from a screen — the patrol a control room does without leaving it.

- **Schedules** name a site, an ordered list of cameras and a set of questions. When its time arrives — in the site's own time zone — a session is created for the assigned officer.
- **A real snapshot at every camera.** The officer works through the cameras, a frame is captured from the live stream at each, and the questions are answered against it. Each snapshot carries a checksum.
- **What was asked is frozen into the session.** Editing a schedule mid-patrol does not change what the officer is being asked, and a patrol from March still reads as it did in March.
- **A failed answer can raise an incident** in the platform's own incident system.
- **One execution, one session** — enforced by the database, so two workers or a restart cannot create it twice. A patrol nobody did is marked missed.
- **A report for every patrol** — PDF and an Excel workbook, stored with a checksum and emailed at once or as a daily, weekly or monthly digest, each period sent once.
- **A Command Centre panel** for patrols due, running and missed.

ADDED SINCE EDITION 1 · MIGRATIONS 0123 TO 0130

15. Drone Patrol

An autonomous drone security patrol built into the platform rather than beside it: the drone's camera is one of the platform's cameras.

- **Fleet, routes, zones and missions.** A drone is assigned to a site and flies a planned route on a schedule or on request, with pre-flight checks, a command queue, and maintenance logs.
- **The same AI watches it.** The detectors that watch fixed cameras watch the drone's. What they see is judged in the context of where the drone was and what that place is, and becomes a drone security event with a risk level. At a privacy area its whole picture is withheld (edition 28).
- **Checked against fixed cameras.** The platform works out which fixed cameras could have seen the same thing, and whether they did.
- **Into the platform's own incident and dispatch.** A serious event becomes an incident, and a guard is sent through the existing dispatch.
- **Verify with drone.** An officer can ask for a flying look at something a fixed camera or a guard reported.
- **Site edge gateway.** A site can keep flying while its link to the platform is down: the gateway buffers what happened and syncs it exactly once when the link returns.
- **A record of every flight** — a PDF and a workbook, emailed on a schedule — with analytics, an analytical risk map and recommendations.
- **Licensed per tenant**, with limits on drones, missions and sites; sixteen permissions of its own; web and phone screens.
- *Documented limitation:* everything here has been proven against a simulator. No physical drone, manufacturer SDK, flight controller or real video stream has been connected.

ADDED SINCE EDITION 1 · MIGRATIONS 0132 TO 0141

16. AI Security Intelligence and human decision support

One flow over every source: events are brought to one shape, related into situations, assessed, and a response is suggested — and then a person decides.

- **One shape for every security event** — CCTV, LPR, face, drone, virtual patrol, alarm, sensor, guard and system events — read from the tables that already hold them, without changing any of them.
- **What a site expects.** Site and camera profiles state what is normal there and when, and planned activity (a work permit, an expected visitor) is read as context.
- **Situations.** Events that belong together — the same plate, the same watchlist entry, neighbouring cameras moments apart — become one situation instead of several alerts.
- **How unusual, and how much it matters.** Each situation is assessed for normality and risk, with the reasons kept beside the score.
- **A suggestion, not an action.** The layer recommends a response and says what the recommendation rests on.
- **A person decides.** The suggestion, the human decision and the action carried out are three separate records. Only a decision made by a signed-in person causes anything to happen, and it happens through the platform's existing functions — dispatch, incident, a drone look.
- **Reports from the ground.** A guard on the phone accepts, arrives and reports what they found; a drone's look comes back as an event and the situation is assessed again.
- **A unified timeline and the evidence** of each situation, with an AI-assisted summary written from templates over recorded facts.
- **A site security score and daily intelligence** — where and in which hours situations begin, repeated vehicles and watchlist entries, and the share that turned out to be false alarms.
- **Feedback.** A person's review of what a closed situation turned out to be is kept, with the suggestion and the decision, as a dataset.
- **Off until asked.** A switch per tenant, off by default. If the background runner is stopped, alerts, incidents and video carry on exactly as before.
- *By design:* the background process cannot act — it reads, records and suggests, and tests hold that line. There is no language model. Nothing learns by itself: a rule or a weight changes only as a tenant setting or released code. A face that matched nobody is *unknown*, never *unauthorised*.

ADDED SINCE EDITION 1 · MIGRATION 0143

17. Smart Investigation

Finding what the platform already knows about a time, a place, a person or a vehicle — and keeping what was found.

- **One search across fourteen sources** — by period, site, camera, kind, event type, severity, risk, plate, name, member of staff and words. Each source is read under its own existing permission and the asker's sites; a source left out is named, with the reason.
- **A typed phrase, read by fixed rules** into that same search. The screen shows which words became which filter and which were not used, so the person can correct it. A phrase with nothing understood is refused rather than guessed at.
- **Investigations.** A file of why somebody was looking and what they found: references to records of any of the fourteen kinds, with notes, in the order they happened, each read live as the reader may see it.
- **The trail of a plate or a watchlist face** — every sighting, in order, with the time and the distance between.
- **Every search is on the record** with what was asked, not what was found.

- *Documented limitation:* an unidentified person is not followed across cameras, and the platform does not claim to. There is no search by a picture of a face.

ADDED SINCE EDITION 1 · MIGRATION 0144

18. Evidence packages and chain of custody

What was kept about a matter: put together, sealed, held against deletion, and accounted for.

- **What belongs is found and offered.** For an incident or an investigation: the frame of that detection, the recording of that camera at that moment, the drone media of that sighting. A person chooses what goes in.
- **Sealing.** A manifest of every item with its checksum is written, and the SHA-256 of the manifest. From then on a database trigger refuses every change.
- **Export** as a ZIP of the manifest as sealed and each original, byte for byte, with its checksum computed again and compared — plus a marked viewing copy of each picture.
- **One chain of custody per package**, from capture to release: captured, collected, accessed, sealed, placed under a hold, exported, downloaded, shared, released, hold lifted — each with who, in what role, and why.
- **Holds.** A hold stops all three retention jobs — frames and clips, recordings, drone media. Sealing places one on every item; a hold is lifted only by a person, with a reason.
- **Removed when due, by a person.** Past the organisation's period, with nothing held and its matter over, a package is listed for Admin or Manager to remove with a reason; its record and chain of custody are kept.
- *Documented limitations:* video is not watermarked; nothing is digitally signed; the platform sends evidence to nobody — sharing and release are records of what a person did.

ADDED SINCE EDITION 1 · MIGRATION 0145

19. The security map

One operational map, and the places of a site as somebody who knows it drew them.

- **Ten layers on one map** — sites, cameras, guards, open incidents, live alerts, open situations, drones, patrol checkpoints, places and drone zones — each shown under its own screen's existing permission and the reader's sites. What has no position is counted under the map, not hidden.
- **The places of a site.** Buildings, floors, gates, access points, emergency and assembly points, zones and parking — a point, an outline or a level of a building, drawn by an administrator. Optional: a site without them works as before. A place is retired, never removed.
- **A door on the map.** An access point can name a door the platform knows, which gives door events a place to appear.
- **What is near.** Selecting an incident, alert or situation lists the cameras, drones, checkpoints and places within a radius, and every guard on shift at the site, nearest first.
- **Where a guard last was.** The last position each guard recorded this shift — a check-in, a scan, a status change — always with its age, and marked stale past an hour.
- *By decision:* guards are not tracked continuously. *Documented limitations:* there are no floor plans; distances are straight lines, not routes; alarm panels and sensors are not drawn.

ADDED SINCE EDITION 1 · MIGRATION 0146

20. Guard response, SLA and escalation

A sending as its own record: what the guard did about it, how long it took, and who is told when it is late.

- **The steps of a response.** Accepted, set off, arrived, reported — or cannot attend, with a reason — each written through to the incident's own status, history and arrival time. A guard who is not coming, or is stood down, gives the incident back.
- **The guard is told on their phone**, and answers there: Accept, On my way, I am there, Report what I found, I cannot attend.
- **Who to send — a suggestion with its reasons.** The guards on shift at the site, ranked by a score made of stated parts: free or already sent, distance by last recorded position and how old that position is, what they were already sent on this shift, and the certifications the site requires. A person chooses and dispatches.
- **Three clocks per incident** — acknowledge, arrive, resolve — judged every minute against the organisation's own settings. A clock that runs out is recorded once, marks the incident as breached, and is told to the person the settings name.
- **Escalation policies.** When an incident is still not acknowledged, reached or resolved after so long, tell a role or a person — by site and severity. Each step is recorded once, with how many people it reached.
- **A response desk** showing what each open incident is waiting for, with the procedure for it beside it.
- *By design:* the clocks are off until an organisation switches them on, and then apply only to incidents opened since. Nothing dispatches, reassigns or re-dispatches by itself.

ADDED SINCE EDITION 1 · MIGRATION 0147

21. The occurrence book: reviewed, corrected and handed on

The book read back — and what one shift hands the next.

- **Search** by words, kind, site, author, shift, period, where its review stands, and whether it has been corrected.
- **Supervisor review.** An entry is noted, marked to be followed up, or its follow-up closed with what was done — by somebody other than who wrote it. Every review is kept. What a reviewer wrote is shown to the people who keep the book, not to a client or a viewer.
- **Correction by a further entry.** A mistake is put right by a new entry that records which entry it corrects and why. The first stays exactly as written.
- **Two more kinds of entry** — a delivery, and unusual activity.
- **Instructions in force at a site.** Issued with or without a date they run out on, read by each guard, closed with a reason. The guards on shift are told, and the instruction is carried into every shift's summary until it ends.
- **A shift's summary, drafted by the platform and confirmed by a person.** Written from what was recorded during the shift — the book, incidents, alerts, patrols, dispatches, visitors, what the site has in hand, instructions, follow-ups — in fixed sentences that count and quote. The guard reads it, corrects it and confirms it; what the platform drafted is kept beside it, and once confirmed it cannot be changed.
- *By design:* the summary writes no prose of its own and draws no conclusion; no language model is involved. An entry is added and never rewritten: no code edits or removes one, and since edition 7 the application's database role is not allowed to either.

ADDED SINCE EDITION 1 · MIGRATION 0148

22. The SOP library

Procedures in versions, approved before they are in force, and found by their own words.

- **A procedure and its versions.** A code it is cited by, a category, a site or every site. Versions are kept: drafted, submitted, approved to be in force from a date, or rejected with a reason.
- **Approved by somebody other than its author.** An approved version cannot be changed afterwards — a database trigger holds it.
- **In force, and running out.** Of the approved versions whose date has come, the latest is in force. A version that has run out leaves nothing in force rather than falling back to an older one.
- **The document as issued** can be attached to a version, kept with its SHA-256.
- **Asking the library** returns the passages of procedures in force that use the words asked — word for word, each with its procedure, its version and who approved it. When no passage uses those words it says so.
- **The procedure beside the incident.** A procedure names the kinds of incident it is for; the ones in force for an incident of that kind at its site are put beside it — on the response desk, on the phone's incident screen, and on the situation screen.
- *By design:* nothing composes an answer and no language model is involved — the platform never invents a procedure. *Documented limitations:* an attached PDF is kept and not read; the search reads English.

ADDED SINCE EDITION 2 · MIGRATION 0149

23. Visitor and contractor authorisation

Who said a visit may happen, for where, for how long and with whom — read at the gate, decided by the host.

- **An authorisation of a visit or a work permit.** Asked for by whoever registers visitors; approved or declined by the host, or by somebody who manages visits when no host is named. A no says why.
- **For a period, and for places.** Valid from and until; for named places of the site map or for the site in general. Where it stands — waiting, valid, run out, never answered — is its answer and the hour, worked out when it is read.
- **Escort and ID.** Whether the visitor is to be escorted, and by whom; and the kind of ID a named person says they saw. The number on an ID is not taken, and the one on the visit is never returned.
- **What the gate reads.** What stands on the record, as sentences made from it. It informs: checking a visitor in is unchanged, and does not look at the authorisation.
- **Cancelled or extended** by the host or somebody who manages visits, each with a reason. The host is told on their phone when asked; whoever asked is told the answer.
- **Where a badge was used.** The door events of the badge a visitor was given, while they had it, are set against the places and the period the visit is authorised for. One outside either is listed for somebody who manages visits to look at, who records that it was in order or how it was followed up.
- **Handed to the intelligence layer, when the organisation asks.** A switch, off unless somebody who may change the organisation's settings turns it on. With it on, a badge used outside its authorisation and not yet looked at becomes an event of the layer — low in severity, naming nobody, carrying the authorisation so that somebody who may read visits can open it — and the layer sets it beside what the cameras and the doors reported there and then. *Added in edition 7.*
- **On the phone:** the visits waiting for your answer, and what stands for a visit beside its check-in.
- *By design:* an authorisation admits nobody and refuses nobody, and a door event outside it is not a finding — nothing is raised and nobody is accused. *Documented limitations:* a visitor is followed only through doors their badge number opened; the people working under a work permit are not followed; with the switch off, which is how it starts, the layer is given none of these events; nothing opens or locks a door.

ADDED SINCE EDITION 2 · MIGRATION 0150

24. Device health, assets and maintenance

The health of devices as they report it, a register of what the organisation owns, and the work that keeps it working.

- **One reading of every device** — camera, recorder, sensor, drone, edge gateway, alarm panel — from what each reports: working, degraded, down, not known or switched off, with why and since when.
- **What is not known is not called working.** A camera with no stream, a recorder not probed for a day, a device that has never reported. A sensor reading a dangerous value is a working sensor.
- **What is kept.** Every five minutes each device's state is read and kept when it has changed: the states it has been in, and how long it was down in the last week and month.
- **The asset register.** A code, a kind, make, model, serial number, where it is, vendor, installed, warranty, status. An asset may be the device it is and then shows that device's health; a UPS, a server or a switch is an asset with no reading, and is shown as that. Known devices are put into the register in one step. Retired, never removed.
- **Work orders.** Raised on an asset, for a facility defect or for a site; given to one of the organisation's people or a named vendor; started; completed with what was done, the parts used and the time out of use; or cancelled with a reason.
- **Preventive schedules.** Something done every so many days puts its work forward before each date.
- **Suggested, then accepted.** A schedule falling due — and, once an organisation asks for it, a device read as down for a set number of hours — puts a work order forward. It is a suggestion until somebody who manages maintenance accepts it or dismisses it with a reason; it assigns nobody and tells nobody.
- *By design:* every answer and screen that shows a reading names what is not measured. *Documented limitations:* frame rate, latency, packet loss, the quality of the picture and gaps in a recording are not measured; an outage shorter than five minutes can pass unrecorded; no cause is inferred; there is no parts store, cost or maintenance SLA; on the phone, whoever an order is given to says it has started and what was done, and nothing else of this is there.

ADDED SINCE EDITION 3 · MIGRATION 0151

25. Risk patterns and advice

Where what went wrong gathers — by hour, day, week and place — what stands out, how much history that rests on, and what a person made of it.

- **Five kinds of thing that went wrong, counted the same way:** incidents; doors refused, forced or tampered with; patrols missed or failed — a guard's tour, a virtual patrol, a drone patrol; devices going down; response clocks missed.
- **Counted three ways,** over the last one to twelve weeks: by weekday and hour where the site is, by week, and by place — a camera, a door, a patrol, a device. Nobody is named.
- **A week of hours** for each kind, shaded in four steps with a key that says in counts what each shade stands for; every cell readable without its shade, and the numbers themselves on request.
- **What stands out, by fixed rules:** a band of four hours, a weekday or a place holding most of the records; twice as many in the later half of the period as in the earlier; a device down three times or more. Each statement has its count in it and one thing to consider.
- **How much history it rests on.** Much, some or little: how many records, over how many weeks, and in how many of those weeks the same held by itself. One busy afternoon is one week, not four.
- **Nothing before is not called a rise.** When the earlier half of a period has no records, the statement says that the counts cannot tell a change from the start of recording.

- **A person's answer.** For one site: accepted, or not accepted with a reason. The platform counts the advice again first and keeps its own statement with the answer. An answer is added and never rewritten.
- *By design:* a pattern that recurred is not a forecast, and every reading says so; confidence is how much history a statement rests on, not a probability; an answer raises no work, moves no guard and alters no roster. *Documented limitations:* no risk score is made; nothing allows for how much is watched at a place, so more records there is not more danger there; a record made while testing is counted like any other; places are not zones; there is no report or export; the phone has no part of it.

ADDED SINCE EDITION 4 · MIGRATIONS 0152 AND 0153

26. Operations board, daily briefing and reports

Every part of the operation counted together — for a site, a customer's sites or every site — a day put into words for a person to publish, and the records taken out as files.

- **Seven sections, counted when somebody asks:** incidents; response; patrols — a guard's tour, a virtual patrol, a drone patrol; guards on shift; devices; visitors; maintenance. For the last 1, 7 or 30 days, with what stands at this moment marked "now".
- **Each section under its own permission.** Holding the board does not show the incidents of somebody who may not read incidents: what is not shown is left out and named. Somebody held to particular sites sees those sites.
- **A time is the middle one of those measured**, with how many it was measured from. When there were none it is nothing, not zero; and response clocks that are switched off are not called "none missed".
- **Site by site, and by customer.** One row for each site, what is at no site, the sites together, and each customer's sites summed — without their times, which are counted for a customer only when the board is asked for that customer.
- **The daily briefing.** One calendar day where the site is, for a site or for every site, drafted as fixed sentences with a count in each, and up to three pieces of the risk advice (module 25) word for word. Each line says what it is true of: the day, the moment of drafting, or the weeks before.
- **A person reviews and publishes.** The reviewer leaves whole sections out — and the briefing says which — and adds a note shown as theirs. No line is rewritten. Once published a briefing is not changed: a correction is a new revision, and the earlier one stays, marked as replaced.
- **Nine reports as CSV files:** the board site by site, response, device health, maintenance, visitor authorisations, door events, risk advice, evidence packages and investigations. Each is a list of records as they are, made when it is asked for and stored nowhere.
- **Taking records out is a permission of its own**, on top of the one the records are read under, and every report taken is audited — who, which, for where, how many. Text a spreadsheet would run as a formula is made plain; each time names its time zone; a file cut at 10,000 records says so. A support session takes none.
- **Emailed, to the organisation's own people only.** A published briefing goes to its list; a report on a schedule goes as CSV and PDF, each person's own sites.
- *By design:* nothing is scored — no index, grade or rank of a site, a customer or a person — and the board and a briefing name nobody. No model writes a briefing, and nothing drafts or publishes one by itself. *Documented limitations:* the board keeps no history of itself; a site with nothing scheduled has nothing missed; the client portal is unchanged; the phone reads published briefings and has nothing of the board or the reports.

ADDED SINCE EDITION 4 · MIGRATION 0154

27. Workforce readings and recommendations

What is recorded of a guard's work and of a site's, read together — each count beside how much there was to do — and what a manager might consider.

- **Six sections from rows the platform already keeps:** shifts; patrols; responses; violations; training and certificates; handovers. For each guard, each site and the sites together, over 7, 28 or 90 days.
- **Not an appraisal.** Each count stands beside how much there was to do — shifts late beside shifts worked, tours missed beside tours due — because a guard given more shifts has more chances to be late.
- **Nothing is scored and nobody is ranked.** There is no total of a person; a list of guards is in order of name, and the table has nothing to sort it by.
- **What a reviewer set aside is not counted against anybody.** A violation that was waived is said to be waived, and one that is disputed is said to be disputed.
- **One's own reading.** A guard, an operator and a supervisor read their own, whole, with what is recommended for them. Somebody else's reading is read by the organisation's own people, and that it was read is written in the audit log.
- **Training, for a guard:** a certificate a rostered shift needs and the guard does not hold; a course whose latest pass has lapsed or lapses within 30 days; three or more assigned tours missed in four weeks, with the courses the library holds — or that it holds none. No course is invented.
- **Cover, for a site compared with itself:** three or more shifts not started; its busiest four hours holding half its incidents and rostered more thinly than the rest of its day; guards arriving half as slowly again at night as in the day.
- **A manager's answer.** Accepted, or not accepted with a reason, kept with the recommendation as it stood, added and never rewritten.
- *By design:* a recommendation is never an employment decision and never a change to a roster — accepting one assigns no course and moves no shift. Lateness and violations are counted and not turned into recommendations: they have their own review, by a person. Fixed rules; no model. *Documented limitations:* nothing says why a shift was not started, and leave is not read; cover is compared from shifts as rostered; a reading is not taken out as a file; nothing is sent to a guard; the phone shows a person their own reading, and nobody else's.

ADDED SINCE EDITION 4 · MIGRATION 0155

28. Security cases

A matter held together: the records it is about, the people working it, what each has to do, who is named in it, and a closing somebody else agrees to.

- **A case** has a number, a title, a kind — theft or loss, trespass, damage, safety, access, other — a priority, a site or no one site, a lead and investigators. A person opens it, from an incident, an investigation or an evidence package, or from nothing.
- **It refers to what it is about; it copies none of it.** Incidents, investigations (module 17) and evidence packages (module 18) are linked by reference and stay where they are. Taking a link off a case removes nothing of theirs.
- **A linked record is read under its own permission and the reader's sites.** A link the reader may not read says that it is one, and what kind, and nothing else.
- **Worked by the people on it** — its lead, its investigators and whoever manages cases. Tasks are given, and end done, with what was done, or dropped, with why. Somebody given a task finishes that task and does nothing else to the case.
- **People and vehicles named in it**, each with how it is connected — reported it, saw or heard it, was affected by it, is named in it — written in by a person.

- **Told, and escalated.** Whoever is put on it is told; an overdue or quiet case reaches Managers.
- **Closing takes two people.** Whoever asks for a case to be closed says what was found; somebody else approves, or declines with why. The screen does not offer approving to whoever asked, the server refuses it, and the database refuses it again.
- **A closed case is not changed** — the database refuses any change but reopening it, with why. Nothing is removed: an investigator, a link or a name is taken off, and the row stays.
- **Its own history** of every step, added to and never rewritten; and **a report** of the whole case in order, on screen and as a PDF, with a record the reader may not read shown as that.
- *By design:* being named in a case is not an accusation — there is no word for a suspect, and nothing names anybody by itself. No rule opens a case. Every step is a signed-in person's and is audited. *Documented limitations:* a name is text, not matched to a visitor, a member of staff, a watchlist entry or a face; closing a case does not resolve its incident or seal its evidence package; escalating changes nothing about a case; the report is not signed; on the phone, somebody given a task says it is done or why it is dropped, and the case itself is not there.

ADDED SINCE EDITION 5 · MIGRATION 0156

29. Data retention and subject reports

How long each kind of record is kept, said in one place as the jobs read it — and where a person appears in the records, when somebody asks.

- **One statement of every retention period in force.** Six kinds of record a job removes — evidence, recordings, drone footage, drone flight tracks, gateway receipts, the audit log — each with its period, where that is set, what removes it and whether a hold stops it.
- **It reads what the jobs read, the same way** — their own functions and constants — so it cannot say one number while a job applies another. A period that falls back to the installation's default says that the job's own is the one in force.
- **Site by site, and the holds.** Each site's own period for its recordings, where its recording policy sets one; how many holds are in force, and how many of each kind of record are held now.
- **No period is a statement too.** Most of what the platform keeps is removed by no job. Everything in modules 17 to 28 is of that kind until an organisation sets a period, and each of its tables is listed with whether it names a person.
- **Twelve periods an organisation may set.** Closed cases and investigations, authorisations that have run out, answers to advice, and since edition 25 guard responses, shift summaries, ended instructions, device health, finished work orders, briefings and evidence packages. Nothing is removed until a period of at least thirty days is set; setting one says first how many are older. A job removes what is over once a day, never what is open — and never a package, which a person removes.
- **A subject report.** For a member of staff, a visitor, or a name or a number plate as it was typed: where they appear in the records of modules 17 to 28, how often and between which dates — told apart as records that concern them and steps they took.
- **Every column that names a person is read.** The list is compared with the database's own catalogue by a test, so a column added later cannot be left out.
- **Who looked for them.** How many investigation searches asked about the person, the plate or the words, and by how many people.
- **Asked by a person of the organisation** — not an API key, not a support session, not somebody held to particular sites — with what is typed sent in the body of the request and never in its address. Each report is one line in the audit log.
- **A sweep behind it.** Every table and route of modules 17 to 29 is asked the same questions by a test, from the database's catalogue and the route table: row level security forced, a permission on every route, every write audited and a person's, each exception named.

- *By design:* the statement says what is configured, not what the law requires, and changes no period; a subject report says where and how often, not what each record says — a record holds other people too; a name typed is matched as text and identifies nobody. *Documented limitations:* settings in use and the chain of custody have no retention period, and the kinds that may have one have none until it is set; a report does not read attendance, rosters, leave, pay, training or violations, or the account, which has its own export; nothing is erased on request; the phone has no part of it.

ADDED IN EDITION 14 · MIGRATIONS 0159 TO 0162

30. Fleet tracking and geofences

The organisation's vehicles, where their trackers say they are, the journeys they made, and what is alerted when they come and go.

- **Vehicles** — registered, changed and archived with a reason, never deleted, and restored; the plate unique in the organisation, compared letters and digits only; department, driver, service status, custom fields and groups. Somebody limited to sites sees their sites' vehicles. Every change is in the audit log.
- **Trackers through Traccar.** The platform's own Traccar runs in the Docker stack when switched on, or an organisation connects its own. Positions are forwarded with a secret, kept once at the device's own fix time, and what is refused is counted with its reason. Tested with the Traccar Client phone app.
- **Online, stale, offline** from the age of the last fix (5 and 30 minutes unless the organisation sets its own), recorded once a change. The live map draws each vehicle at its last fix; the live connection says which vehicles moved, never where.
- **Journeys and playback** worked out when asked: trips, stops and silences over up to seven days, inaccurate fixes and impossible jumps left out and counted, played back fix by fix; exports are audited. Positions and archived vehicles are kept until the organisation sets a period.
- **Geofences** — circles, polygons or a site's boundary, for a site's vehicles or chosen groups. A fix wobbling on the line changes nothing: it must be 15 m past it, or its own accuracy, and held for a second fix or 30 seconds. Rules for entering, leaving, staying and speed, each with its severity and the people told, raise alerts in the one alert queue; a vehicle in service that stops reporting raises one alert an episode.
- **What else the site recorded** around an entry or exit — plate reads, barrier commands, door events and open incidents, each to whoever may read it — said to be "plate matches" or "near in time only".
- *By design:* a geofence never opens a barrier, admits anybody or confirms who is driving, and nothing is called confirmed: a plate that matches is not who was at the wheel. *Documented limitations:* only Traccar, and no hardware tracker has been tried yet; the phone reads the fleet's status and draws no map; the older position endpoint checks geofences its old way; the default map tiles are not for a map left open all day.

ADDED IN EDITION 15 · MIGRATIONS 0163 TO 0166

31. Footage Review — uploaded video

Video that did not come from a live camera — a client's export, a phone recording, or a saved recording — searched by description, with events and possible altercations found in it. Nothing found raises an alert or an incident: it waits for a person.

- **Uploads in 16 MB parts** that carry on where they stopped: MP4, MOV, MKV and AVI, up to 4 GB and 6 hours, to disk or S3.
- **Kept 30 days**, then removed with everything worked out from it, each removal audited; an evidence hold keeps a video until it is lifted. A saved recording can be copied in, masked as it was recorded.
- **Search by description.** SigLIP 2 (Google, Apache-2.0), run inside the platform, turns one frame a second and the typed words into vectors; the nearest frames are ranked by the model's own probability, said as strong, possible or weak. Nothing below weak is offered, a search with no strong or possible match says so, and every search is audited with its wording.

- **Events** — the live workers' person and vehicle detector counts people and vehicles in each sampled second; the seconds with some are events on the video's timeline. A count is how many, never who.
- **Possible altercations**, in two stages: frames compared with descriptions of fighting and of its look-alikes (hugging, dancing, a crowd walking); where that holds and two people are there, a pose model looks five times a second for people within reach and a fast arm. Never an assault, never a name: a person marks each confirmed or not, audited.
- **Its own worker** — ONNX Runtime and ffmpeg, no PyTorch, every model in its image and checked against published checksums; it reaches nothing on the internet, and only the API reaches it. A chart, GCC values, S3 settings and a controls sheet make the deployment package.
- *Documented limitations:* the search and the altercation check are not yet measured on real footage, and every threshold is a first setting; the worker is not a core service; a real S3 bucket, reached with the pods' role since edition 16, has not been tried; the detectors are AGPL-3.0, and SaaS use needs Ultralytics' licence; the package has never been run.

ADDED IN EDITION 18 · MIGRATIONS 0170 TO 0174

32. The AI assistant and learning

An assistant that reads and works the whole application as the person using it, and twelve features that learn from the organisation's own work. A licence of its own: the AI suggests and prepares, and a person decides.

- **Its own licence**, granted per organisation by the platform owner: the features, until when, the assistant's model and a monthly reply limit.
- **The organisation's admin** switches each feature on, and chooses the roles, the sites and how sure a warning must be.
- **The assistant** on the web, the desktop app and the phone. Ask in your own language and it replies in it. It reaches the application's own functions, about 1,100, with the person's own sign-in, so their permissions, their sites and the audit log apply as when they click.
- **A change waits on a Confirm card** showing exactly what will be sent. Nothing changes until the person presses Confirm, and it is audited as theirs. Passwords never pass through the model. Deleting an organisation, finalising payroll, deleting evidence, changing a role and switching off two-factor sign-in are never offered: they stay manual.
- **Two models, chosen per organisation by the platform owner:** a free one on the organisation's own server (Ollama), or Claude (Anthropic), paid per reply and held to the limit. NRIC/FIN, passport and bank numbers reach the hosted model shortened to their last four characters.
- **A new staff member from their IC.** The front and back are read on the platform's own server (RapidOCR) and the photos are not kept; the number is checked against its check letter; the person corrects every field; the assistant asks the rest one question at a time. The full number never reaches either model.
- **Learning from the organisation's own work:** late and no-show warnings, overtime and fatigue, false alarms, device failures, roster suggestions, missed patrols, best patrol times, guard work trends, an incident helper and a payroll check. Simple models fitted to its own records, refitted about once a day, each warning with the factors that raised it, measured on history it did not see; the bar it must reach moves with what people accept and dismiss.
- **Decisions pop up** beside the AI button on every screen, one at a time, and a phone is pushed when warnings are raised. Conversations are kept; each person reads only their own.
- *By design:* nothing in the module acts on its own. A warning changes no roster, alert or payslip, and the assistant saves nothing without Confirm. *Documented limitations:* the development PC runs a small free model (1.5B), helped to answer 12 of 15 everyday questions right; it can still state a detail wrongly, so real use wants the 7B one on a server, or the hosted one. The phone's assistant has not run on a device; the IC reader is tested on printed cards only; a feature warns only with enough history.

ADDED SINCE EDITION 1 · MIGRATIONS 0102 TO 0114

33. Platform operations — the vendor's console

Seventh AI sells this to security companies, so the vendor and the customer are different businesses and do not share a login, a permission set or a bill.

- **A platform tenant of its own.** The platform owner works from the vendor's tenant, not from inside a customer's, and holds none of a customer's day-to-day permissions — rosters, payroll, incidents, evidence. A customer's command centre is the customer's.
- **The customer's lifecycle** — trial, pending, active, suspended, expired, cancelled — rather than on or off, with a grace period after a subscription ends.
- **Pricing and licensing.** A catalogue of what each module costs and how it is charged, tied to the module licences the AI workers enforce. Since edition 7 it lists the fourteen modules built after it was written — Virtual Patrolling, security intelligence, investigation, evidence packages, the security map, guard response, the SOP library, visitor authorisation, assets and maintenance, risk advice, the operations board, workforce readings, cases and data retention — so the vendor can switch each on or off for a customer, as it could the first twelve. A module nobody has switched off is on with its product, so no customer lost anything by it. The catalogue is the vendor's record of what a customer has been sold: it does not by itself stop a screen from answering — only the AI modules' licences and Drone Patrol's do that.
- **Drone Patrol's own licence.** Manage Licenses has a tab for it, one organisation at a time: on or off, an expiry, and how many drones, missions and sites — a limit left empty is no limit — with a note. The screen says whether the organisation may use Drone Patrol now, and why not when it may not. *Added in edition 7.*
- **The vendor's own invoices**, for customers who pay by bank transfer rather than by card.
- **Subscription notices** — a trial ending, a payment overdue, a licence about to lapse — each raised once, not nightly.
- **An error centre.** The same failure seen four thousand times is one row with a count: how many distinct things are broken, which customers each touches, since when, and whether anyone has dealt with it.
- **Support sessions, read-only by default.** The operator can look into a customer's account to answer a question without being able to change anything; elevated access is a separate, recorded choice.
- **Platform health and analytics** — recording, the intelligence runner, drone patrol; growth, adoption and churn from nightly snapshots.
- **The owner's account needs more than a password.** Two-factor authentication is enforced for the account that can reach every customer.
- *By design:* suspending a customer for non-payment is off by default. Cutting off a security company's cameras is a decision somebody makes deliberately, with their name on it.

Where AI is used, and where a person decides

"AI" covers several different things in this platform, and they are not trusted equally. This table says, for each place a model or a rule does work, what does it and what stays a person's. **It is the part of this document that is updated whenever an AI capability is added or upgraded** — with what the new capability does, what it is not trusted to do, and what a person still decides.

Where	What does the work today	What a person does
Detection on camera and drone video (11 modules)	Computer-vision models on frames: object detectors, a face embedding model, plate reading. **Edition 8:** a fixed camera's frames reach the models with its privacy zones painted out, so nothing under one is detected, matched or kept	Reads the alert; acknowledges, escalates or dismisses it
Face and plate watchlists	Embedding similarity in the database (pgvector); plate text matching	Keeps the watchlists; judges a match
Roster auto-scheduler	Scores candidates against nine stated rules. Not a learned model	Reviews the draft, edits it, publishes it
Security intelligence (module 16)	Rules, versioned as a named engine: relate events into situations, score how unusual and how serious, suggest a response with its reasons	Decides. Only a person's decision causes an action
Drone events (module 15)	Context rules and a risk engine over what the drone's camera detected; correlation with fixed cameras	Decides what becomes an incident; asks for a drone look
Who to send (module 20)	A score made of stated parts, shown with the suggestion	Chooses the guard and dispatches
A typed investigation phrase (module 17)	Fixed-rule parsing into a structured search, showing which words became which filter	Corrects the search; reads the results
Shift summary (module 21) and situation summary (module 16)	Templates: fixed sentences over recorded counts and quotations	Reads, corrects and confirms it
Finding a procedure (module 22)	Word search over approved text; passages quoted with their source	Reads the procedure and acts on it
A visitor's badge against what was authorised (module 23)	A comparison: the door events of the badge with the places and the period of the authorisation. Not a model. **Edition 7:** when an organisation switches it on, one outside its authorisation is given to security intelligence as an event that names nobody, and is related and assessed like any other	Looks at each one listed; records that it was in order or how it was followed up. Decides what follows from a situation it is part of
Device health (module 24)	Rules over what each device reports: a stream's state, a probe, a heartbeat, a late reading. Nothing is inferred about the cause	Reads it; decides what is work
Work orders put forward (module 24)	Two rules: a schedule falling due, and — when an organisation has switched it on — a device read as down for a set number of hours	Accepts the suggestion as work, or dismisses it with a reason

Where	What does the work today	What a person does
Risk patterns and advice (module 25)	Fixed rules over counts of what was recorded: a band of hours, a weekday or a place holding most of the records; twice as many as before. Each says how much history it rests on. Not a model, and not a forecast	Reads it; answers it for a site — accepted, or not accepted with a reason. The answer changes nothing else
The daily briefing (module 26)	Templates: fixed sentences, each with a count in it, over one day's counts. The same figures give the same lines. Not a model	Leaves sections out, adds a note of their own, and publishes it — or does not. No line is rewritten
Workforce recommendations (module 27)	Fixed rules over four weeks' records: a certificate a rostered shift needs, a course lapsed or lapsing, tours missed, shifts not started, busy hours rostered thinly, slower arrival at night. Not a model	A manager answers — accepted, or not accepted with a reason. The course is assigned and the roster changed by a person, as before
Cases (module 28)	Nothing. No rule opens a case, links a record to one or names anybody in it	Opens it, works it, and closes it — which takes two people
The retention statement and a subject report (module 29)	Nothing decides. The statement reads what the jobs read; a report counts where a person appears. A name typed is matched as text. Not a model	Sets a period for the kinds that may have one, or leaves them kept; removes an evidence package that is due; judges what of a record is that person's
Daily intelligence and the site security score	Counting over a stated period of history	Reads it as history, not as a forecast
Searching uploaded video by description (module 31)	**Edition 15:** SigLIP 2, an image-and-text model inside the platform, ranks frames by likeness to the typed words, its probability said in words. It writes nothing; nothing leaves the platform	Judges each moment by watching it; nothing follows from a match
People and vehicles in uploaded video (module 31)	The person and vehicle detector, one frame a second; seconds joined into events. Counts, never identities	Watches the events; places an evidence hold on a video that matters
Possible altercations in uploaded video (module 31)	The image-and-text model screens for fighting against its look-alikes; a pose model looks for people within reach and a fast arm. Not yet measured	Marks each one confirmed or not an altercation. No alert, no incident
The AI assistant (module 32)	**Edition 18:** a language model (Claude, or a free one on the organisation's server) reads the application as the person and writes the answer. A change becomes a Confirm card. **Edition 19:** a small model is handed the few functions that fit, read first	Presses Confirm or Cancel. Nothing changes without it
Reading an IC (module 32)	Text recognition on the platform's own server; the photos are not kept	Checks and corrects every field
Learning from the organisation's work (module 32)	Simple models fitted to its own records; each warning names its factors, and its bar moves with people's decisions	Accepts or dismisses each warning, on AI Insights or the card that pops up, and changes nothing by it

What the platform does not do today

- **A language model answers only in the assistant (module 32)**, and it saves nothing by itself: a change waits for the person's Confirm. Only the hosted model is outside the platform, and only where the platform owner chose it. Elsewhere a typed question is parsed, not understood, and a summary is a template; Footage Review's SigLIP 2 (module 31) only ranks frames.
- **Nothing decides for a person.** No part of the platform dispatches a guard, opens or closes an incident, flies a drone to an unplanned place, operates a door, admits or refuses a visitor, raises maintenance work, publishes a briefing, assigns a course, opens or closes a case, or changes a roster on a recommendation alone. No record of modules 17 to 28 is removed unless a person set the period for it.
- **Learning is stated and bounded.** Only module 32's features learn: each fits a simple model to the organisation's own records and moves its bar by people's decisions, within stated limits. Elsewhere a rule or a weight changes only as a tenant setting or as released code.
- **Nobody is accused by a rule.** A face that matched nobody is *unknown*. An unidentified person is not followed across cameras. A prediction says what it rests on: a count, or the factors behind it. A reading of a guard's work scores and ranks nobody, and being named in a case is not an accusation.

In progress and planned

The enterprise expansion was fourteen phases, built and merged one at a time, and all fourteen are built. Phases 1 to 13 are modules 17 to 29 above. Phase 14 added no module: it tests eleven chains end to end, from what a camera saw to a case's report, and holds that every step that mattered was a person's. Nothing is in progress. Privacy zones, found to be recorded and not applied, were decided and built the same day (module 02). The GPS fleet upgrade (14 October 2026) and Footage Review (20 October 2026) were six phases each, every phase merged after its tests passed; all are built (modules 30 and 31).

Decided by the owner on 9 October 2026, and carried out

Each of these was found while building and left as it was, because putting it right changes something that worked. They were put to the owner, who decided that all seven be done.

What	As it was	As it is now
The occurrence book	No code edited or removed an entry, but the application's database role was allowed to	The right is taken away: the database refuses it too
The existing CSV exports	Text that begins with =, +, - or @ was written as it is, and a spreadsheet ran it as a formula	Such text is written as text. Numbers are untouched
The existing data-subject export	Its evidence was the organisation's latest, whoever it showed; it could include where files are kept; it left no audit line	It lists what its subject handled, never says where a file is kept, and is one line in the audit log
Default retention periods	Two were given to the jobs' services and not to the API, so the statement said the job's own was the one in force	The installation gives the API the same two. An installation put together by hand can still differ, and the statement says so
Privacy masks	Read without a token by anybody who knew a camera's id	Read with a credential that may read cameras, within the caller's own organisation — and, since edition 8, applied (module 02)
Retention for modules 17 to 28	Nothing removed a case, a visitor's authorisation or an answer about a guard	Twelve kinds may be given a period by the organisation; nothing is removed until one is set, and a package only by a person (module 29)
Visitor and door events	Set against the authorisation; not given to the intelligence layer	Given to it when an organisation switches that on; off to begin with (module 23)

Waiting on something outside the code

- **A phone build and a device pass.** The phone parts of guard response, handover notes, procedures and visitor authorisation, the four screens added in edition 7, privacy zones on the phone (edition 9) and the AI assistant with its corner button (editions 18 and 20) are type-checked and tested against the server, but have not run on a device since the app moved to Expo SDK 57. Guards have them after a new build.
- **A code-signing certificate.** Desktop 1.0.15 is built and carries the AI module. Its installer is not code-signed, so Windows warns before it runs - and where Smart App Control is on, may refuse a newly built installer for some minutes before it allows it.
- **Real hardware.** A physical drone, access-control controllers and alarm panels have not been connected.
- **A cluster install.** The Helm chart lints and renders in CI and has not yet been installed on a cluster.
- **Footage Review (module 31).** A first run on a real S3 bucket; Ultralytics' licence for SaaS; and footage of real altercations, to measure the check against.
- **The assistant's model (module 32):** a server for the 7B free one, or a Claude key.

Client applications

Application	Platform	Scale	Notes
Web console	Browser (React)	124 routes	The full platform. Every module, every administrative surface.
Mobile app	iOS and Android (React Native, Expo SDK 57)	59 screens	Built for the guard on shift: check-in, patrol scanning, occurrence book, handover, post orders, incidents and the response to them, situations, drone events, SOS — and, for whoever is signed in, their own reading, published briefings, and the work orders and case tasks given to them, and the AI assistant. Whoever manages privacy draws and deletes a camera's privacy zones. Offline outbox queues actions taken with no signal and sends them when it returns.
Windows desktop	Electron	Same as web	Packaged installer for control-room machines, with auto-update. Version 1.0.15, which installs over 1.0.14. It is a copy of the web console taken when it is built: a fix to the web reaches it in the next build.

How the web console moves, waits and fails

One set of rules for all 124 routes, kept in one place in the code, so the same thing looks and reads the same wherever it happens. What each page has is listed page by page in `PAGE_AWARE_ANIMATION_MATRIX.md`, which is written from the pages and checked against them by a test.

- **Every page has its own entrance and its own loading.** As a page is opened, a drawing of what it is about makes itself over the page's name - camera tiles lighting up for the live wall, a sweep for a map, rotors for a drone screen, a bell for alerts - and clears in about half a second; if the page is still fetching, the same drawing goes on, small, beside the page's own words until the first answers are in. Twenty-four drawings for the pages inside the console.
- **It never takes the screen from the person using it.** The page beneath answers a click or a key from its first moment and the first of either ends the entrance. Man down, emergency alert, alarm panels, the response desk, the action center and a drone patrol in flight are never covered. Each person can switch page animations off, or all motion down, in Settings; the device's own reduced-motion setting is followed.
- **A page that could not fetch says so.** Every page that fetches has four answers and not three: a placeholder shaped like the content while it loads, what failed and a way to try again when it fails, "nothing here" only when the server answered with nothing, and the content. Before, a failed request read as an empty list.
- **A changed filter keeps the rows that were there**, dimmed, behind a line that says the list is being refreshed, on 47 lists. One record is never kept under another's name.
- **What arrives is marked and said.** On eight screens that refresh themselves - alerts, incidents, drone events, situations, the response desk, man down, the action center and the command centre's alert feed - a row or card that was not there before is marked in its severity's colour for six seconds, and a screen reader is told how many.
- **A camera's tile says what it is doing**: connecting, buffering, or no picture with a way to try again - on the live wall, its full-screen view, a situation's live view and the drone screens. An alert on a camera is a red frame that beats three times and stays.
- **Nothing is drawn as live that is not.** A vehicle's marker pulses only while it is moving and was heard from in the last two minutes; a drone in flight says "Live" or "Stale" beside its telemetry and is drawn faint once it has

not been heard from for twenty seconds. Both go by the clock, not by when the page last fetched.

- **Said when it is so.** "Snapshot saved" on a patrol when the server has the snapshot - it completes nothing; "Export ready" when the browser has the file. No percentage is shown that the server did not report.
- **Limits.** The motion has been checked stopped at a moment, not watched at a screen's frame rate. A frame-by-frame camera tile whose stream stops part-way keeps its last frame and says nothing: the browser reports nothing. Cards that are clicked but are not buttons cannot yet be reached by keyboard.
- **The Windows app** has all of this since 1.0.8; **the phone** keeps the same rules (tested, not yet on a phone).

Roles

Eight built-in roles ship with the platform; tenants may define their own in addition. A role cannot grant a permission it does not itself hold. The permissions of modules 16 to 29 are held by the roles that work a site — not by Super Admin, and not by the Client role.

Role	Typical holder	Scope
Super Admin	Platform operator (the vendor)	The platform: customers, pricing, licences, errors, support sessions. Not a customer's operations
Admin	Agency owner or operations director	Everything within one tenant
Manager	Operations manager	Multi-site oversight and approvals
Supervisor	Site or shift supervisor	Rostering, attendance, review and incident response
Operator	Control room operator	Live monitoring, alerts, decisions and dispatch
Security Guard	Officer on shift	Mobile app: own shifts, patrols, responses and reports
Viewer	Auditor or observer	Read-only
Client	The agency's customer	Their own sites, incidents and invoices

Built for Singapore

- **Digital Occurrence Book** built against the Private Security Industry Act requirement for a licensed agency to maintain one. No screen or endpoint edits or removes an entry; a mistake is corrected by a further entry.
- **CPF** from CPF Board's published tables by date, with PRs' graduated rates and the Additional Wage ceiling.
- **IR8A** annual reporting.
- **PWM grades** across the seven-grade Progressive Wage Model ladder, with wage floors that pay is checked against, and **PLRD licence** number and expiry per officer.
- **What a site requires an officer to hold**, with a daily report of who does not meet it.
- **Gazetted public holidays** with holiday pay, and the 72-hour monthly overtime cap flagged on every payroll run.
- **SGD invoicing** with purchase-order and payment-terms fields.
- **Work pass tracking** with expiry warnings.
- **PDPA** consent and data subject request handling; identity numbers kept to the minimum each record needs.
- **Asia/Singapore** as the tenant timezone, applied to every rendered date.
- **Four languages** reflecting the local workforce: English, Chinese, Malay and Tamil.

Known limits

Stated here in one place, as well as beside each module, because a reference that leaves them out is a brochure.

- **Drone Patrol has flown only in a simulator.**
- **Access control and alarm panels** have no real hardware behind them on the development system.
- **The assistant's small test model can state a detail wrongly** (12 of 15 right). Footage Review's search and altercation check are **not yet measured** on real footage.
- **An unidentified person cannot be followed across cameras.** A plate can; a watchlist entry can.
- **Guards are not tracked continuously.** A guard's position is the last one they recorded, shown with its age.
- **Device health is what devices report.** Stream state, disconnections, probes, heartbeats and late readings are read; frame rate, latency, packet loss, the quality of the picture and gaps in a recording are not measured, and an outage shorter than five minutes can pass unrecorded.
- **A privacy zone is fixed to the picture, not to the scene**, and takes effect within about ten seconds: what was captured before it is unchanged. A camera with one has no HLS view; the AI is blind inside it, so a restricted zone that overlaps it is too; a drone's picture is withheld whole near a privacy area, judged by where the drone is, not where it looks; and a recorder at the site that records the camera itself is outside it. It has run on test streams, not on a customer's camera.
- **A visitor is followed only through the doors their badge opened**, and those door events reach the intelligence layer only where an organisation has switched that on.
- **A work order put forward by the platform** says what was read, not what is wrong: no cause is inferred for a device that is down.
- **Risk patterns are counts over history**, not forecasts. No risk score is made, nothing allows for how much is watched at a place, and a record made while testing is counted like any other.
- **The operations board is counts of what is recorded**, and keeps no history of itself: how things stood at an earlier moment is not known. A briefing is drafted when a person asks and is sent to nobody; a report is a CSV file, on no schedule.
- **A workforce reading is not an appraisal.** Nothing says why a shift was not started, leave is not read, and cover is compared from shifts as rostered.
- **A case names nobody by itself**, and a name in one is text: it is not matched to a visitor, a member of staff, a watchlist entry or a face.
- **A subject report says where a person appears, not what each record says**, and reads modules 17 to 28: attendance, rosters, leave, pay, training and violations are each on their own screen.
- **The end-to-end tests ran on software only.** No camera, door controller, alarm panel or drone was behind them, and none was timed or loaded.
- **Payroll:** CPF's 2027 rates are not loaded; their employer and employee shares are not yet published.
- **Evidence:** video is not watermarked and nothing is digitally signed.
- **The phone app** needs a pass on a device; **the desktop app** is not code-signed.
- **A signed-in browser tab left open keeps the web console it loaded** until it is reloaded. The sign-in page fetches a newer console by itself.
- **The Helm chart** has not been installed on a cluster.

Compiled from the Seventh AI Vision source repository at migration 0179 — 179 database migrations, 319 tables, 125 API modules serving 1,169 operations, 154 backend services, 124 web routes and 59 phone screens, covered by 7,052 automated tests (backend 4,369, repository inspection 1,406, web 840, phone 437). Capability counts reflect what is merged and running, not what is in progress or planned.